

OLS Diagnostics

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Goals

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- ➊ What assumptions do we make in OLS?
- ➋ How can we tell if we violate our assumptions about OLS?
- ➌ What is multicollinearity?

Regression Assumption #1: Linearity

The population regression function is linear in the parameters:

- Violation of the linearity assumption:

$$Y_i = \frac{1}{\beta_0 + \beta_1 X_i} + \varepsilon_i$$

- Not a violation of the linearity assumption:

$$Y_i = \beta_0 + \beta_1 X_i^2 + \varepsilon_i$$

- We talked about how to allow for non-linearities in X in regression.

Regression Assumption #2: Random Sample

We should have a iid random sample of size n .

Violations:

- Time-series
- Selected samples (i.e. observed, convenience)
- Spatial dependence

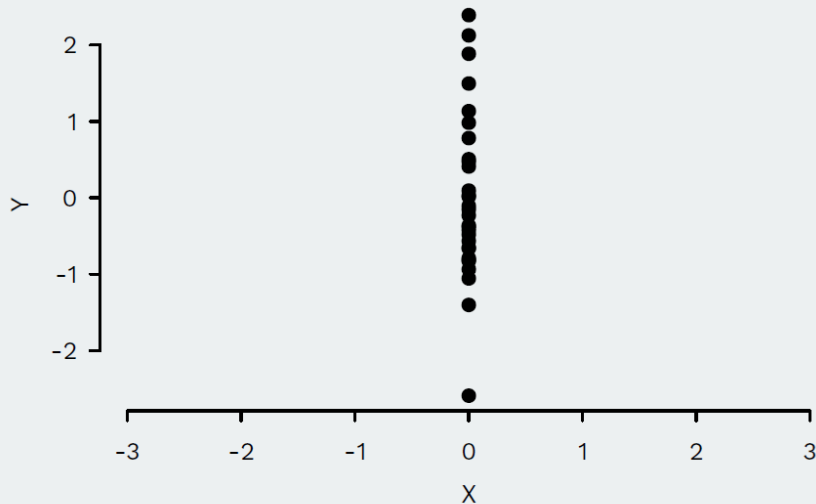
#3: Variation in X

- OLS not well-defined if no in-sample variation in X
- Remember the formula for the OLS slope estimator:

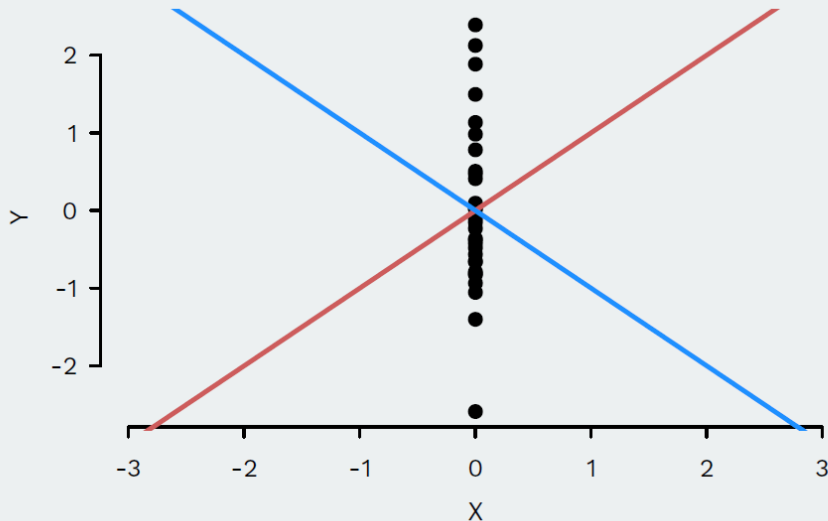
$$B_1 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

- What happens here when X_i doesn't vary?

Variance in X



Variance in X



#4: Zero conditional mean

- The error, ε_i , has expected value of 0 given any value of the independent variable
- When is this assumption most plausible? When X_i is randomly assigned.
- What if we are using observed data? We must control for factors that explain both Y and X .

Properties of OLS Estimator

Under a certain set of assumptions, the OLS estimator is:

- Consistent: as the sample size grows large, our estimate of beta converges (in probability) on its true value.
- Efficient: it converges on its true value more quickly than other estimators as N increases.
- Unbiased: estimate approximates the truth on average.

Gauss-Markov Assumptions

OLS regression is the Best Linear Unbiased Estimator (BLUE) when:

- Errors have mean zero.
- Errors are uncorrelated.
 - ▶ A random shock on one observation tells us nothing about a random shock on another observation.
 - ▶ There are no left-out variables that cause y and are correlated with the other x variables.
- Errors have equal variance for all values of x : homoskedastic and not heteroskedastic.
- Errors are normally distributed conditional on the model.

Assumption of Regression

We violate these assumptions when:

- The true relationship between our variables is not linear (errors will not be normally distributed and will not have equal variance).
- Dependent variable is discrete (errors cannot be normally distributed).
- Extreme outliers exist (errors do not have constant variance).

OLS Robustness

- How do we know if we violate the Gaussian assumptions of OLS?
- Do we meet the Gauss-Markov assumptions?
 - ▶ Is our dependent variable conditionally normally?
 - ▶ Are any patterns left in our residuals after we have specified our model? If so, why?
 - ▶ Or are our residuals a normally distributed “cloud” of points around the regression line?
- How can we check?
- Look at the residuals!!!

Residuals

- Model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ where $\epsilon_i \sim N(0, \sigma^2)$.
- Observed outcome for i is Y_i at predictor value X_i
- Predicted outcome for i is $\hat{Y}_i$ at predictor value X_i is

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

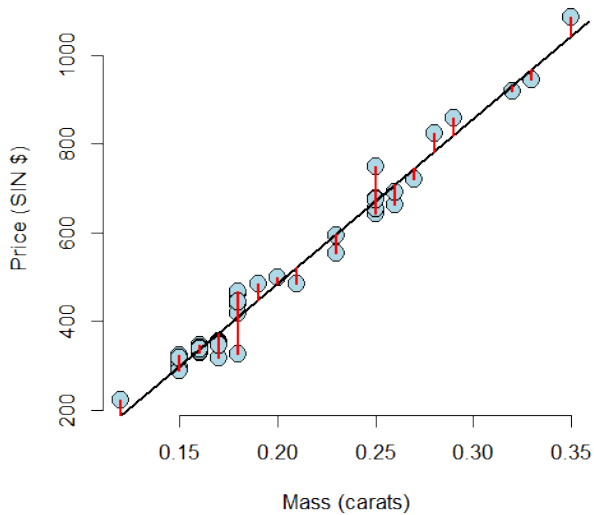
- Residual, the difference between the observed and predicted outcome

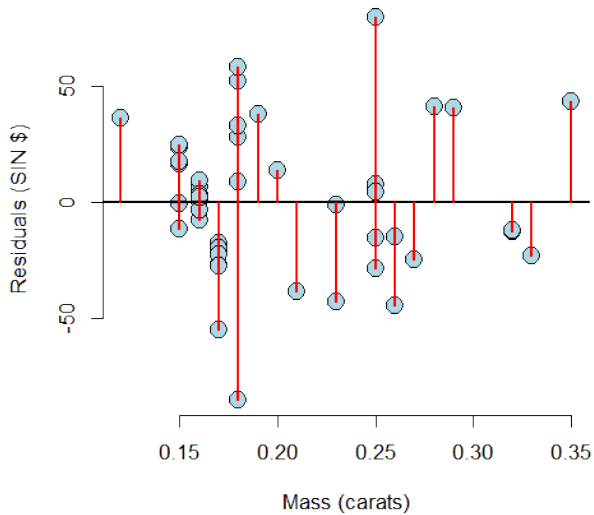
$$e_i = Y_i - \hat{Y}_i$$

- The vertical distance between the observed data point and the regression line
- Least squares minimizes $\sum_{i=1}^n e_i^2$
- The e_i can be thought of as estimates of the ϵ_i .

Properties of the residuals

- $E[e_i] = 0$.
- Residuals are useful for investigating poor model fit.
- Positive residuals are above the line, negative residuals are below.
- One differentiates residual variation (variation after removing the predictor) from systematic variation (variation explained by the regression model).
- Residual plots highlight poor model fit.



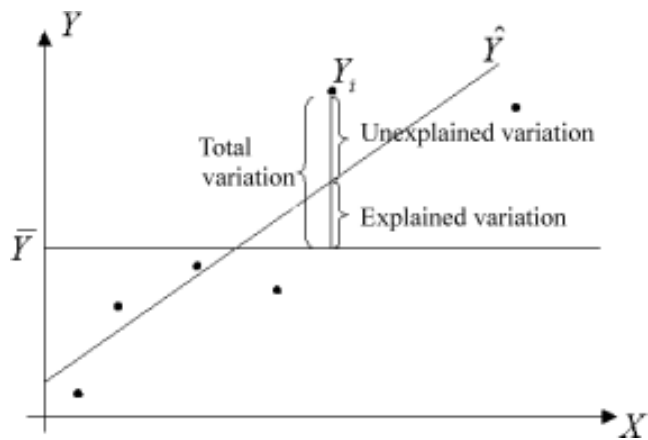


Estimating residual variation

- The total variability in our response is the variability around an intercept (think mean only regression) $\sum_{i=1}^n (Y_i - \bar{Y})^2$
- The regression variability is the variability that is explained by adding the predictor $\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$
- The error variability is what's leftover around the regression line $\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$



$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$



Outliers, Leverage, and Influence

- Leverage (Potential) — an observation has high leverage if it has an x -value far from $\bar{x}$.
- Influence — an observation has influence if its deletion leads to a large change in the fit.
- Observations with high leverage may or may not have influence.
- We cannot just look at magnitude of residuals to determine leverage and influence.

Outliers

Calling a point an outlier is vague.

- Outliers can be the result of spurious or real processes.
- Outliers can have varying degrees of influence.
- Outliers can conform to the regression relationship (i.e being marginally outlying in X or Y, but not outlying given the regression relationship).

How do I use diagnostic tools

- Be wary of simplistic rules for diagnostic plots and measures. The use of these tools is context specific. It's better to understand what they are trying to accomplish and use them judiciously.
- Not all of the measures have meaningful absolute scales. You can look at them relative to the values across the data.
- They probe your data in different ways to diagnose different problems.

How do I use diagnostic tools?

- Patterns in your residual plots generally indicate some poor aspect of model fit. These can include:
 - ▶ Heteroskedasticity (non constant variance).
 - ▶ Missing model terms.
 - ▶ Temporal patterns (plot residuals versus collection order).
 - ▶ Residual QQ plots investigate normality of the errors.
 - ▶ Leverage measures (hat values) can be useful for diagnosing data entry errors.
 - ▶ Influence measures get to the bottom line, 'how does deleting or including this point impact a particular aspect of the model'.

Regression Diagnostics

Cook's Distance

- Cook's distance estimates the influence of individual cases on the fit by removing observations one at a time, and comparing the fit when the observation is removed to when it is present.

- $$D_i = \frac{\sum_{j=1}^n (\hat{Y}_j - \hat{Y}_{j(i)})^2}{p * MSE}$$

- ▶ where p is the number of fitted parameters in the model and MSE is mean squared error.

- As a rule of thumb, $D > 1$ (or sometimes $D > 4/n$) is large.

Regression Diagnostics

Diagnostic Plots

- Residual vs. Fit Plots: residuals on y-axis and $\hat{y}$ on x-axis
- Studentized Residual Plots: rescales residuals so they follow a t-distribution
- hat-values: measure of leverage
- Normal density plots
- Q-Q plots: plot quantiles of residual distribution vs normal distribution.

Omitted Variable Bias

- What happens if we run a bivariate regression when the true relationship is multivariate?
- If the two independent variables are correlated and both cause y , leaving one out of the regression will give us a biased estimate for the variable we do include in the model.
- The effect of the left out variable will be attributed to the included variable to the extent that the two are correlated.
- They are confounded and our estimates of β are biased.

Omitted Variable Bias

- Omitting variables results in bias in the coefficients of interest - unless their regressors are uncorrelated with the omitted ones.
- This is why we randomize treatments, it attempts to uncorrelate our treatment indicator with variables that we don't have to put in the model.
- (If there's too many unobserved confounding variables, even randomization won't help you.)
- Including variables that we shouldn't have increases standard errors of the regression variables.

Omitted Variable Bias

- Actually, including any new variables increases (actual, not estimated) standard errors of other regressors. So we don't want to idly throw variables into the model.
- The model must tend toward perfect fit as the number of non-redundant regressors approaches n .
- R^2 increases monotonically as more regressors are included.
- The SSE decreases monotonically as more regressors are included.

Omitted Variable Bias solutions

- Add omitted variable (thanks...).
- But variable might be unmeasurable (or we don't know it's omitted)
- Note that bias is a theoretical construct because we don't observe the true population parameter.
- No definitive test of omitted variable bias, though Stata has associated tests (ovtest, linktest)

Multicollinearity

- If two variables are highly collinear (correlated) then we will not be able to tell which variable is causing the dependent variable.
- The estimates of the coefficients are unstable. Neither variable may be statistically significant, even when both have a true effect on y .
- Bias vs. efficiency trade-off.
- Variables may be jointly significant!

Multicollinearity consequences

- Estimates remain unbiased
- Standard errors will increase
- T-stats will fall (and p-value increase)
- Estimates become sensitive to specification changes.

Standard errors

Remember that of sampling variability, our estimate may be too high or too low. The standard error gives us an indication of how much the point estimate is likely to vary from the corresponding population parameter.

$$se_{bk} = \sqrt{\frac{(1 - R_Y^2) * s_y}{(1 - R_{Xkj}^2)(N - K - 1) * s_{xk}}}$$

Multicollinearity detection

- Correlation coefficients
- Variance Inflation Factors (VIFs) - extent of which a variable is explained by covariates.

$$VIF(\hat{B}_1) = \frac{1}{1 - R_i^2}$$

- Rule of thumb: $VIF(\hat{B}_1) > 5$ is problem.

Multicollinearity solution

- Do nothing (trade off with omitted variable bias is too great)
- Drop a redundant variable
- Find more data

Tension between OVB and MC

What to do?

- Check for potential multicollinearity.
- Are you omitting any variables that might cause y and be correlated with the other independent variables?
- How do our estimates change as we include and exclude variables?